NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA, SURATHKAL

COMPUTER NETWORKS LABORATORY

NS-3 SIMULATION EXERCISES

LABORATORY EXERCISE SHEET

Instructions

1. Use the NS-3 version installed in your lab environment. Some helper names and configuration details may vary between versions.

2. Read the complete problem statement before beginning the simulation.

3. Submit the simulation script, relevant output or trace files, and a brief explanation of the results.

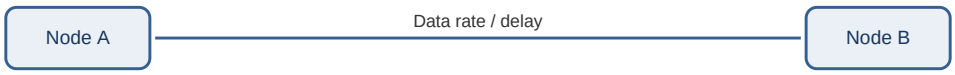
4. Keep simulation duration and traffic settings consistent when comparing scenarios. Include units with all measured values.

5. Label diagrams, address tables, graphs, and output clearly.

Exercise Questions

Question 1. Point-to-Point Link: Throughput and Delay

Problem statement: Create two nodes connected by a point-to-point link. Send UDP traffic from Node A to Node B.



Tasks:

• Run the simulation with a link rate of 5 Mbps, then 10 Mbps. Keep propagation delay unchanged.

• Measure received throughput and average end-to-end delay using FlowMonitor or tracing.

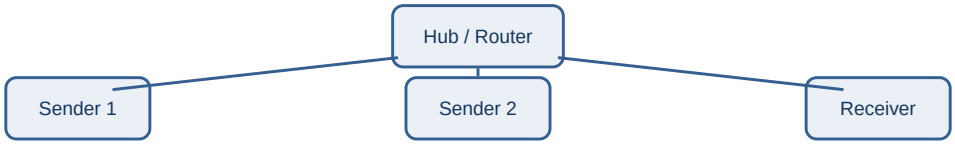
• Briefly explain how changing the link rate affected the results.

Concepts: Point-to-point links, UDP, throughput, delay

Submission: Simulation script and a results table for both runs.

Question 2. Star Network: TCP and UDP

Problem statement: Create a central hub/router node connected to three end nodes to form a star-shaped topology.



Tasks:

• Set up one TCP flow and one UDP flow toward a receiving application.

• Record bytes received or throughput for each flow.

• Explain one difference between TCP and UDP that is reflected in your results.

Concepts: Topology, TCP, UDP, application traffic

Submission: Script, topology description, results, and a short comparison.

Question 3. Packet Loss at a Bottleneck

Problem statement: Connect two senders to a router and connect the router to one receiver. Make the final link slower than the sender links.

Tasks:

• Run with a 10 Mbps bottleneck and then with a 1 Mbps bottleneck.

• Generate traffic from both senders in each run.

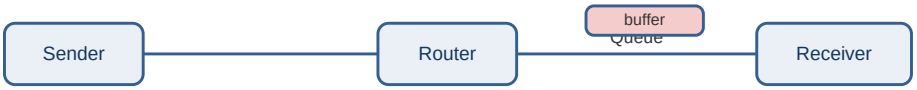
• Compare throughput and packet loss; explain why the slower link may become congested.

Concepts: Bottlenecks, congestion, packet loss, throughput

Submission: Results for both runs and a short explanation.

Question 4. Queue Discipline Comparison

Problem statement: Create a sender-router-receiver path. Configure the outgoing queue with DropTail and one other queue discipline supported by your NS-3 version.



Tasks:

• Keep topology and traffic identical between runs.

• Offer enough traffic to build a queue.

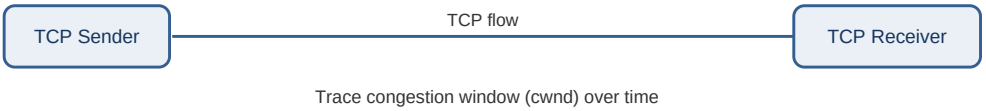
• Compare packet drops and delay, then state which queue worked better for this traffic and why.

Concepts: Queue disciplines, buffering, delay, packet drops

Submission: Configuration details, comparison table, and a conclusion based on measurements.

Question 5. TCP Congestion Window

Problem statement: Create a TCP sender and receiver connected through a point-to-point link. Enable tracing for the TCP congestion window (cwnd).



Tasks:

• Run a TCP transfer long enough to observe cwnd changes.

• Save the trace to a text file.

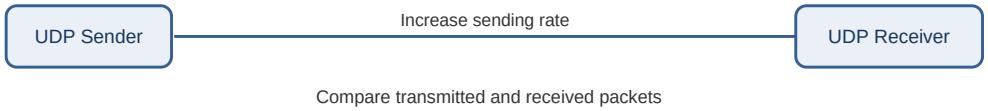
• Plot cwnd against time or tabulate at least five trace entries. Explain two visible changes.

Concepts: TCP, congestion control, cwnd, tracing

Submission: Trace file and a plot/table with a short interpretation.

Question 6. UDP Sending Rate and Packet Delivery

Problem statement: Send constant-bit-rate UDP traffic over a point-to-point link using an NS-3 traffic application such as OnOffApplication.



Tasks:

• Run with an offered rate below link capacity.

• Run again with a sending rate above link capacity.

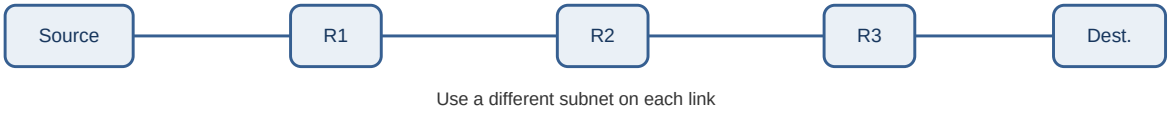
• Compare sent packets, received packets, and packet delivery ratio: received ÷ sent × 100%.

Concepts: UDP, offered load, link capacity, packet delivery ratio

Submission: Results for both runs and the packet delivery ratio calculation.

Question 7. IPv4 Addressing and Multi-Hop Routing

Problem statement: Create a chain of three routers between a source and destination. Assign a separate IPv4 subnet to each link and configure routing.



Tasks:

• Verify that traffic reaches the destination.

• Record the subnet and IP addresses used on each link.

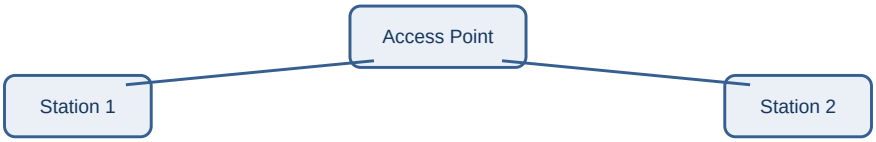
• Intentionally remove or change one route, observe the failure, and restore the correct setting.

Concepts: IPv4 addressing, subnetting, routing, troubleshooting

Submission: Addressing table, connectivity evidence, and a fault/fix explanation.

Question 8. Wi-Fi Performance: Two Stations

Problem statement: Create a Wi-Fi network with one access point and two stations using an NS-3 Wi-Fi helper and a supported standard.



Tasks:

• Send UDP traffic from both stations to a receiving application endpoint.

• Measure aggregate throughput and packet delivery ratio.

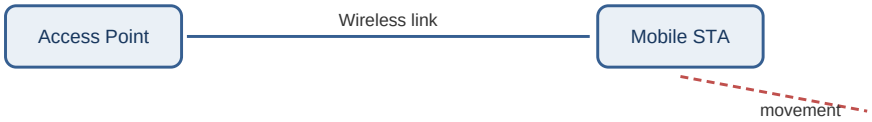
• Change one setting, such as offered load or station position if mobility is configured, and compare results.

Concepts: Wi-Fi, access point, stations, UDP, performance

Submission: Topology/configuration summary, results for two scenarios, and explanation.

Question 9. Mobility and Wireless Connectivity

Problem statement: Create a Wi-Fi network with one access point and one mobile station. Apply a mobility model supported by your NS-3 version.



Tasks:

• Run once with the station near the access point.

• Run again with the station moving farther away along a simple path.

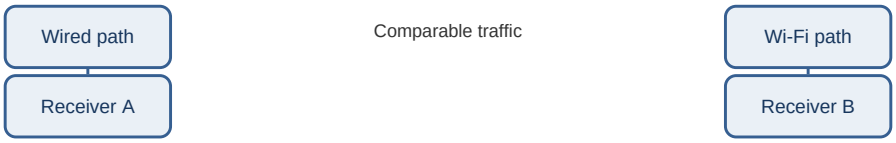
• Compare packet delivery or throughput and explain any observed change.

Concepts: Mobility, wireless connectivity, packet delivery, throughput

Submission: Mobility settings, results for both runs, and a short interpretation.

Question 10. Wired Versus Wi-Fi Performance

Problem statement: Build two comparable scenarios: a wired point-to-point path and a Wi-Fi path. Send similar UDP traffic through each.



Tasks:

• Use the same simulation duration and similar offered traffic rate.

• Measure throughput, packet delivery ratio, and average delay for each scenario.

• Give two possible reasons for any differences in the measured results.

Concepts: Wired and wireless networking, UDP, delay, throughput

Submission: A comparison table and a short conclusion supported by measurements.